

Unit 3: Rapid Prototyping and Manufacturing

When the bell rings, be prepared to start class by:



1. Be seated

2. Cell phone in pouch

3. Backpack stowed

4. Have Fusion 360 on your computer and start relevant work

CNC Mill Operation



Download these files:

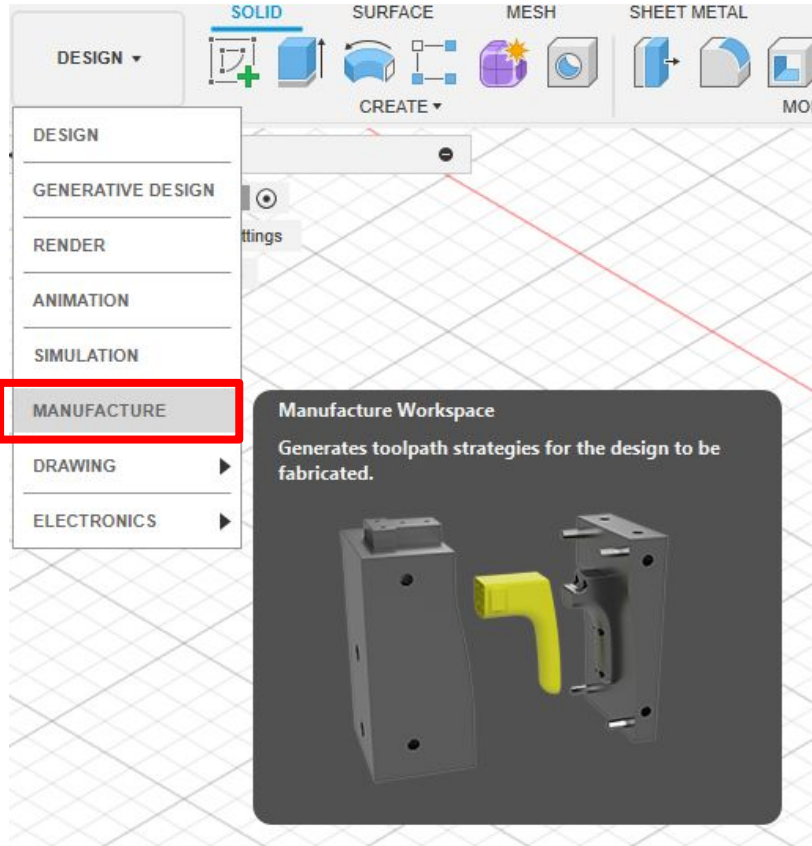
[OMIO Machine](#)

[Crossfire Pro Machine](#)

[OMIO post file](#)

[Crossfire post file](#)

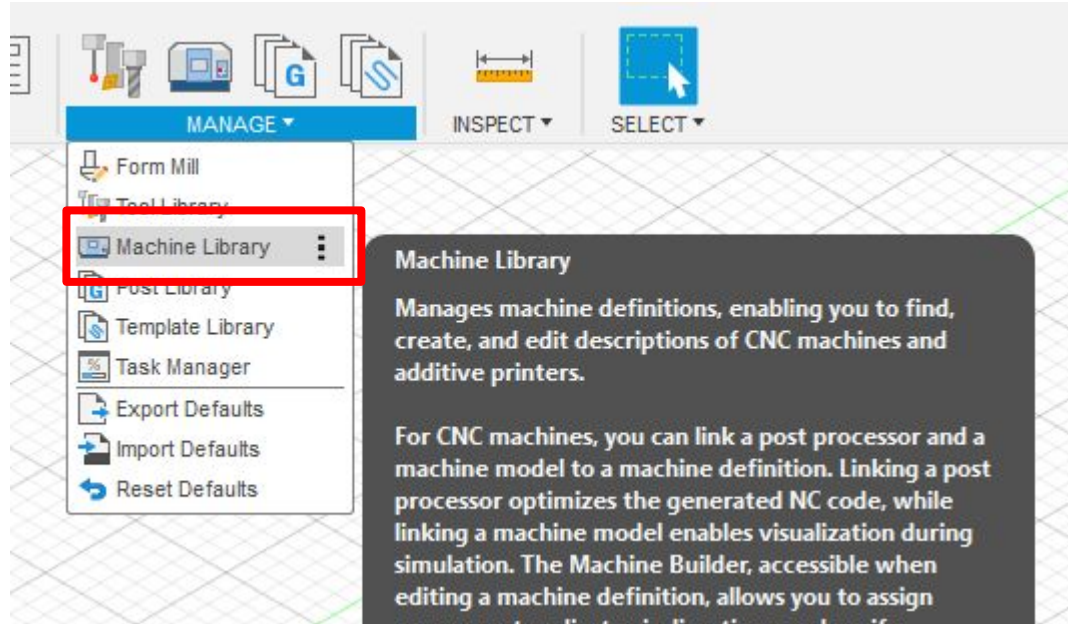
Fusion Manufacturing Setup - Manufacture Workspace



Switch to the
manufacture
workspace



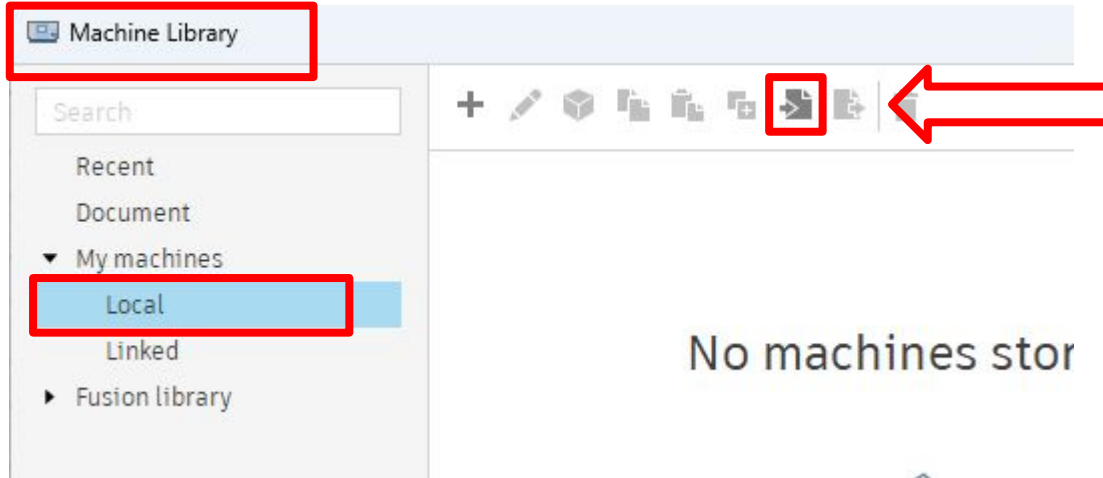
CNC Mill Operation - Machine Library Setup



Add the OMIO CNC Router and the Crossfire Pro Plasma CNC.

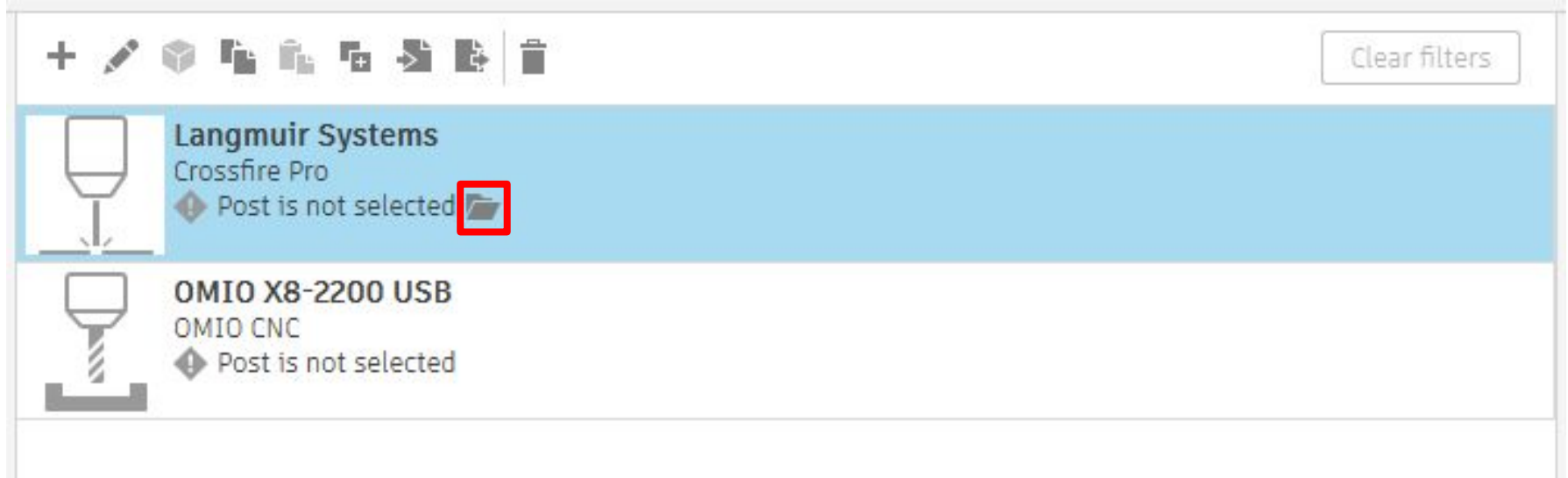
Start by going to the machine library

CNC Mill Operation - Machine Library Setup



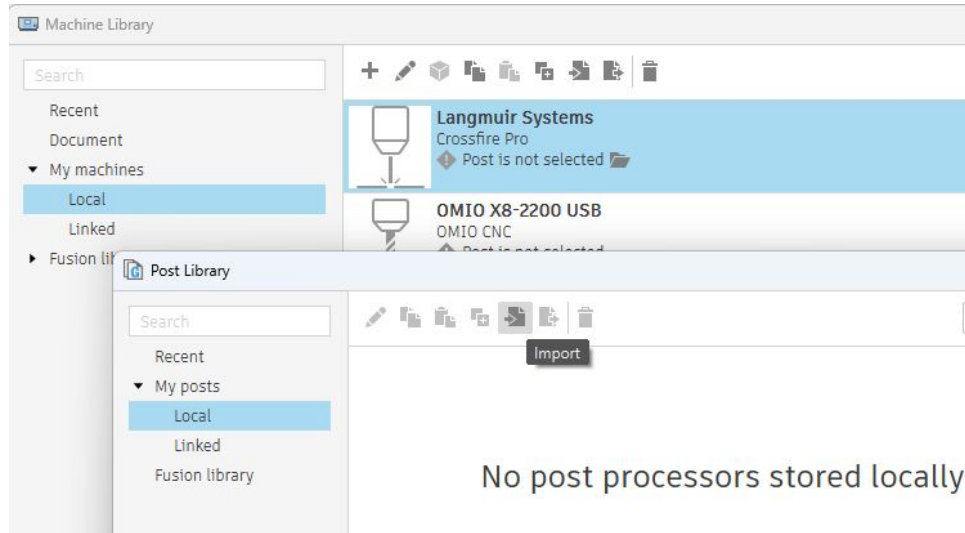
Click on the import icon, navigate to your downloads, and select the .machine file for each machine

CNC Mill Operation - Machine Library Setup



Click on the file folder button that appears when you hover over the machine.

CNC Mill Operation - Machine Library Setup



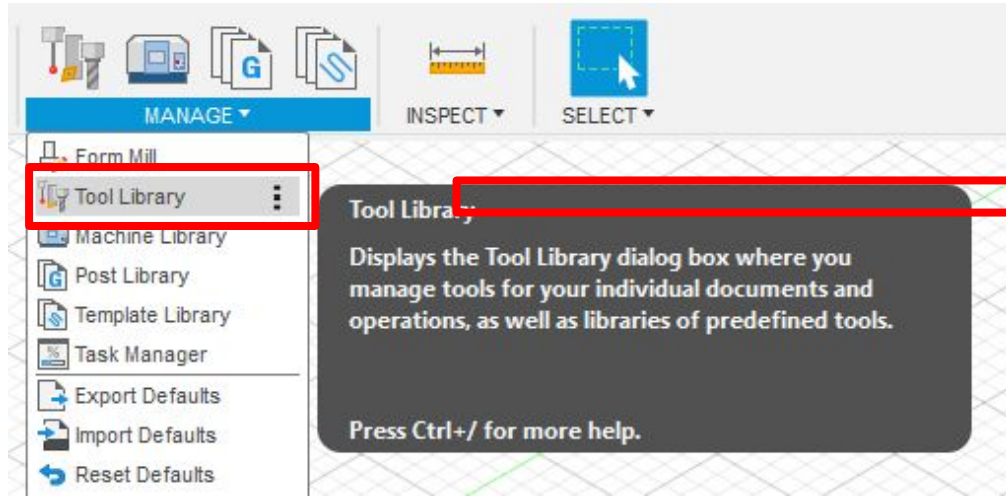
Use the import option to navigate to your downloads folder and select the post file for each machine (fire control for the plasma CNC, OMIO for the OMIO CNC).

CNC Mill Operation - Machine Library Setup

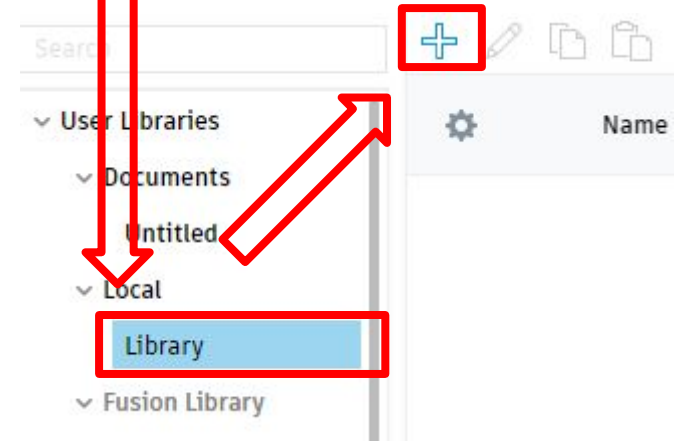


When you are properly setup, your machine library should have these two tools with their associated post file.

CNC Mill Operation - Tool Library Setup



Finally you need to create tools. Open your tool library, click on “Library” under “Local.” Then click “+”



Fusion Manufacturing Setup - Create Plasma Tool

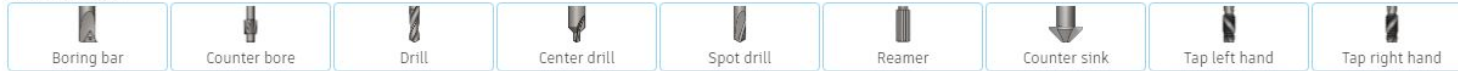
New tool



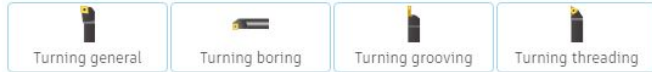
Milling



Hole making



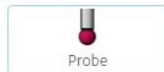
Turning



Cutting



Probe



Holders



Select the Plasma Cutter
tool type

Fusion Manufacturing Setup - Create Plasma Tool

Library /  1 - Plasma cutter

General

Cutter

Holder

Cutting data

Post processor

Description

Plasma Cutter

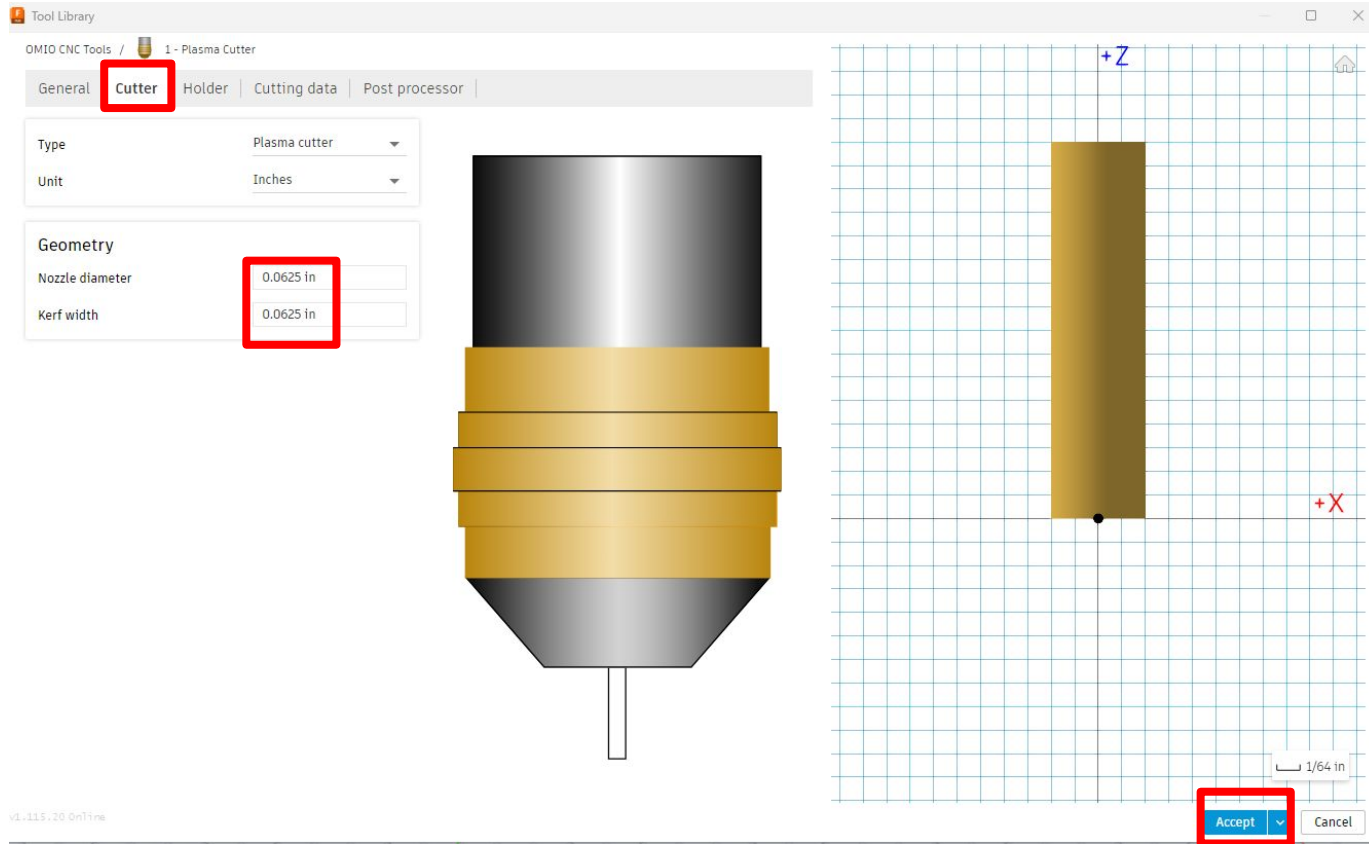
Vendor

Product id

Product link

Give your plasma cutter tool a description

Fusion Manufacturing Setup - Create Plasma Tool



Switch to the **cutter** tab, change the nozzle and kerf dimensions to 0.0625in.

Click “Accept”

Fusion Manufacturing Setup - Create End Mill Tool

While still in the local tool library, add another tool. Select Flat end mill for the tool type.



Fusion Manufacturing Setup - Create End Mill Tool

 Tool Library

Library /  1 - Ø6mm (Flat end mill)

General | Cutter | Shaft | Holder | Cutting data | Post processor

Description	4mm 3-Flute Flat End Mill
Vendor	
Product id	
Product link	

Give your End Mill tool a description

Fusion Manufacturing Setup - Create End Mill Tool

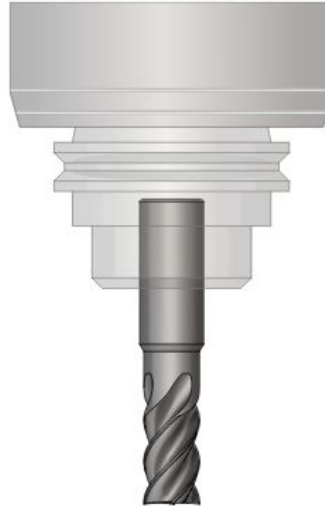
General **Cutter** Shaft Holder Cutting data Post processor

Type Flat end mill ▼
Unit Millimeters ▼
Clockwise spindle rotation ☒
Number of flutes 3
Material HSS ▼

Geometry
Diameter 4 mm
Shaft diameter 4 mm f_x
Overall length 63 mm
Length below holder 20 mm
Shoulder length 14 mm
Flute length 14 mm

Tool Assembly
Gauge length 20 mm f_x

Clockwise Spindle Rotation



In the Cutter tab, modify the highlighted fields. Fields with the “fx” will calculate on their own.

Fusion Manufacturing Setup - Create End Mill Tool

Cutting data | Post processor

Speed

Spindle speed

Surface speed f_x

Ramp spindle speed f_x

Feedrates

Cutting feedrate

Feed per tooth f_x

Lead-in feedrate f_x

Lead-out feedrate f_x

Transition feedrate f_x

Ramp feedrate

Ramp angle

Vertical feedrates

Plunge feedrate

Plunge feed per revolution f_x

Passes and linking

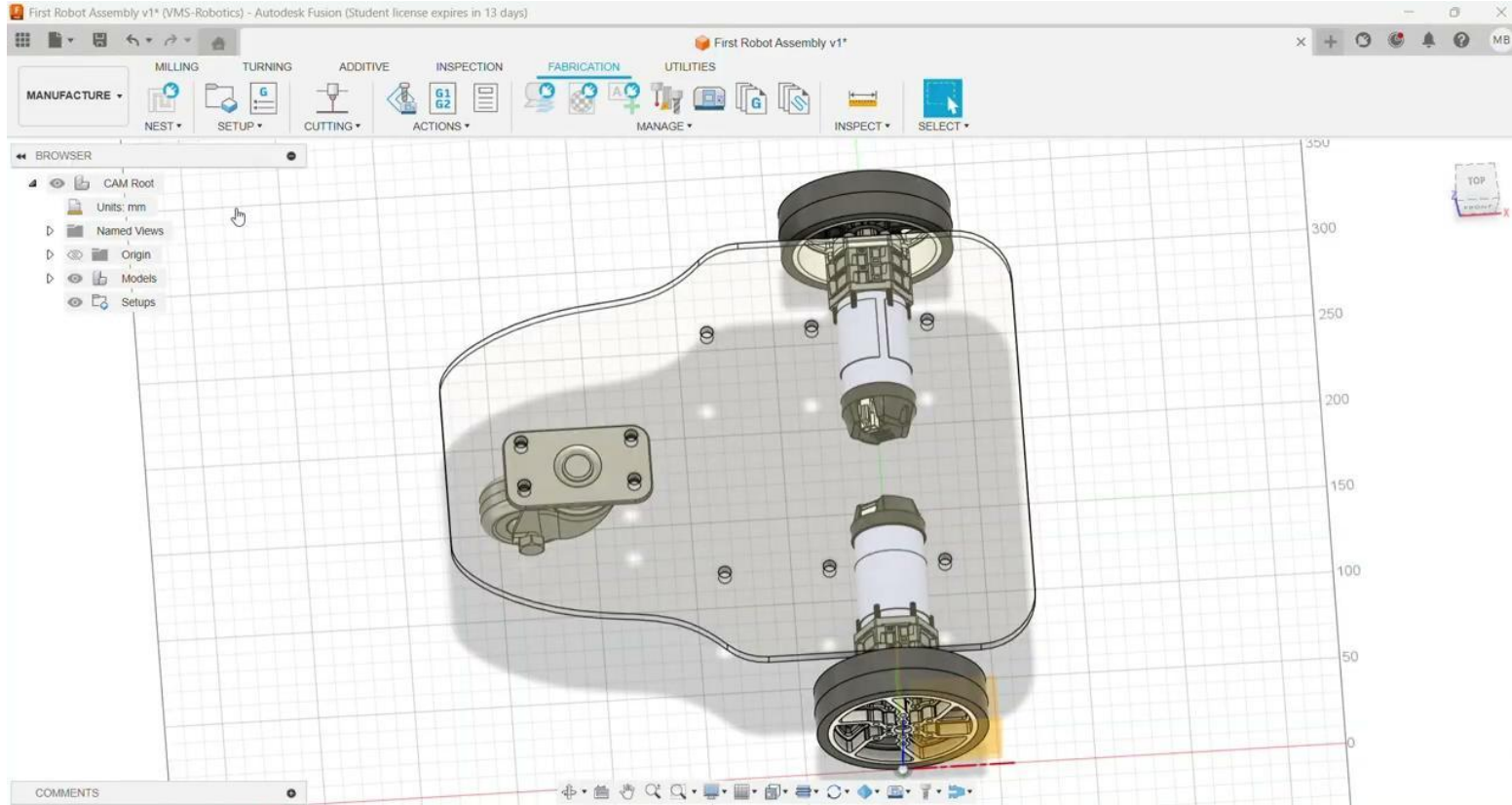
Use stepdown ☐

Use stepover ☐

Coolant

In the cutting data tab, modify the highlighted fields and then click accept.

Fusion Manufacturing Setup - Create Tool Paths



Fusion 360 - Setting up Hubs for Group Work

VMS-Robotics / First Robot Project

Files/Folders in Current Project

New folder Upload Members and permissions Project details

☐	Name		Owner	Size	Date Modified	↓
☐	First Robot Assembly		Matt Bombich		10/29/2025 6:44 AM	
☐	REV Ultraplanetary Motor 2 Cartridges (1)		Matt Bombich		10/29/2025 5:24 AM	
☐	Rev 90mm Grip Wheel (1)		Matt Bombich		10/9/2025 12:15 PM	
☐	2in swivel caster (1)		Matt Bombich		10/9/2025 12:15 PM	

Go to your home screen, click the project you want to share, then click on Members and Permissions. This takes you to the Autodesk website

Fusion 360 - Setting up Hubs for Group Work

🏠 > First Robot Project

[Content](#) [Members and Permissions \(2\)](#) [Wiki Pages](#)

You can assign permissions at the project, folder, or subfolder level. While members with access to a specific folder can see the entire folder structure it belongs to, members can only see the data within their assigned folder. [Learn More](#)

Add or search members and groups

  Add Member  Create group

☐ Name	↑	Permission Level
 Admins (Admin Group)		
 Matt Bombich		

Enter the emails for the classmates you wish to add (i.e.
11257@vicksburgschools.org)

Fusion 360 - Setting up Hubs for Group Work

🏠 > First Robot Project

[Content](#) [Members and Permissions \(2\)](#) [Wiki Pages](#)

You can assign permissions at the project, folder, or subfolder level. While members with access to a specific folder can see the entire folder structure it belongs to, members can only see the data within their assigned folder. [Learn More](#)

Add or search members and groups

🔍 Type emails or group names, then press Enter ↵ or click on 'Add'



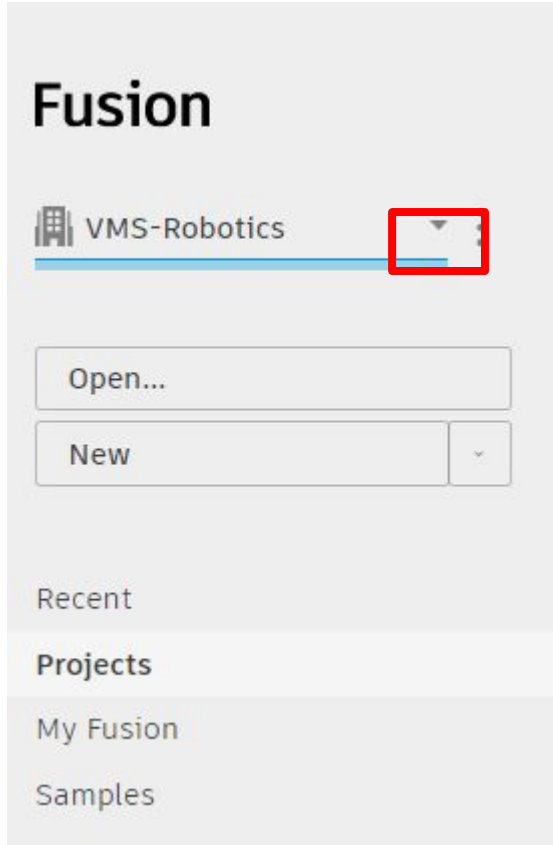
👤 Add Member

👥 Create group

☐ Name	Permission Level
👤 Admins (Admin Group)	
👤 MB Matt Bombich	

Click on the permission level for each member and assign them editor permissions.

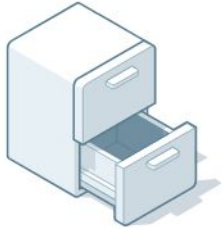
Fusion 360 - Setting up Hubs for Group Work



Invitees: In Fusion 360, from your home screen, change to your new hub and navigate to your shared project. From there you may upload and modify content.

Don't FREAK OUT if you don't see the new hub or the content in the new hub right away. It takes some time to update the server.

Fusion 360 - Setting up Hubs for Group Work



There is nothing here yet.
To get started, create or open a document.

[Learn More](#)



Don't FREAK OUT if you don't see the new hub or the content in the new hub right away. It takes some time to update the server.

CNC Mill Operation

- **Personal Safety:**
 - Required PPE: Safety glasses, close-toed shoes.
 - Prohibited: Loose clothing, untied hair, jewelry, gloves near blade.
- **Pre-Operation Checks:**
 - Ensure your work is not in conflict with work at the adjacent tools
 - Ensure work surface is clear of debris.
 - There must be sufficient stock remaining to hold it firmly in place while cutting.
 - Turn on cooling water pump
 - Check blade for damage
- **Operation:**
 - Keep hands at least 6 inches from moving parts.
 - Do not leave machine running unattended.
 - Fasten material, make measurements or adjustments only when machine is stopped.
- **Post-Operation:**
 - Turn off machine and cooling water pump.
 - Clean work area of sawdust and debris.
 - Store tools and materials properly.

CNC Mill Operation

CNC Mill True/False Test (10 Questions)

1. Safety glasses are not required for CNC machine operation. (F)
2. Untied hair is acceptable if it's short. (F)
3. The work area must be clear of debris before starting. (T)
4. Small work pieces can be held by hand during CNC operation.
(F)
5. The emergency stop should be inaccessible during use. (F)
6. The machine can run unattended during long cuts. (F)
7. Measurements can be taken while the spindle is moving. (F)
8. Proper feed and speed settings are optional. (F)
9. After use, the work area should be cleaned of chips. (T)
10. Remote control must be returned to position post-operation. (T)

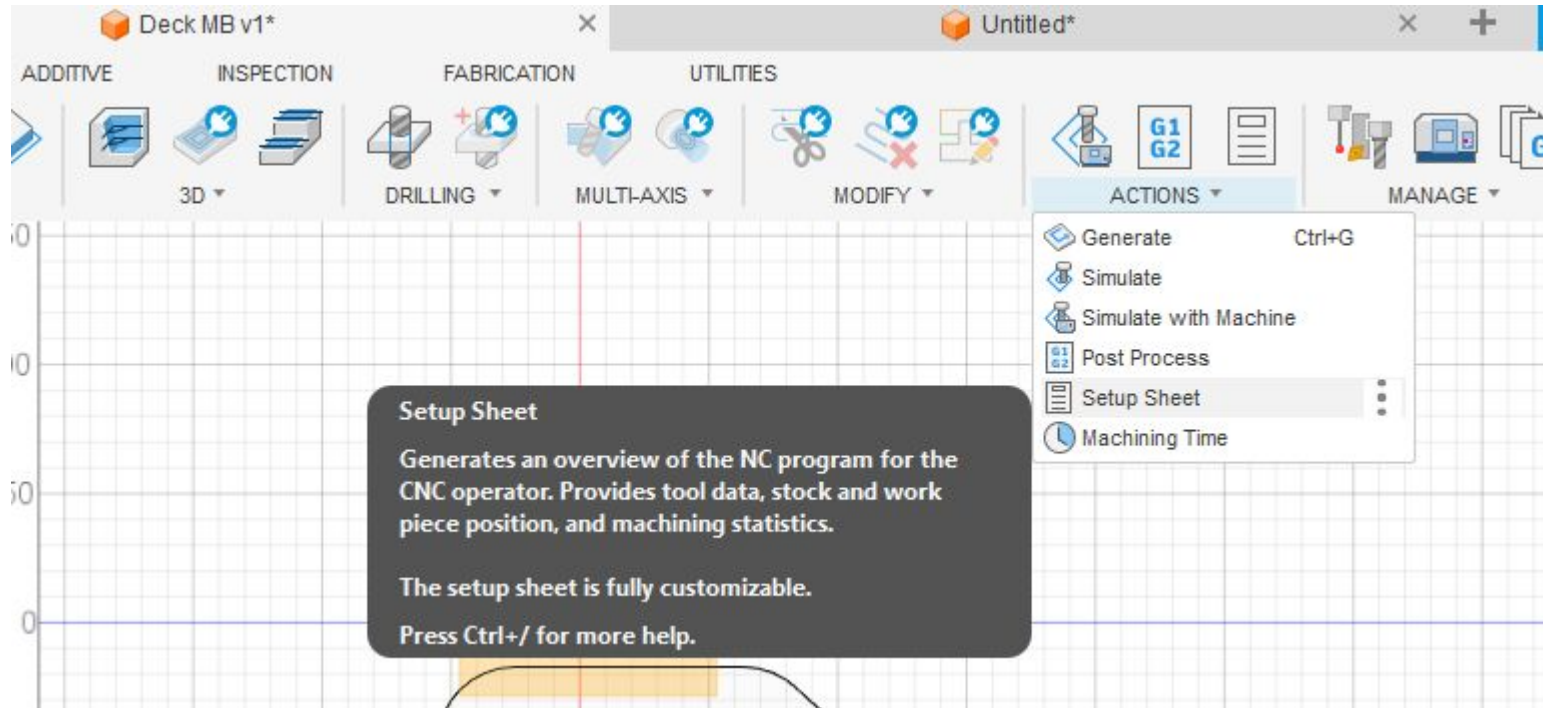
CNC Mill Operation



Radial Arm Saw Operation

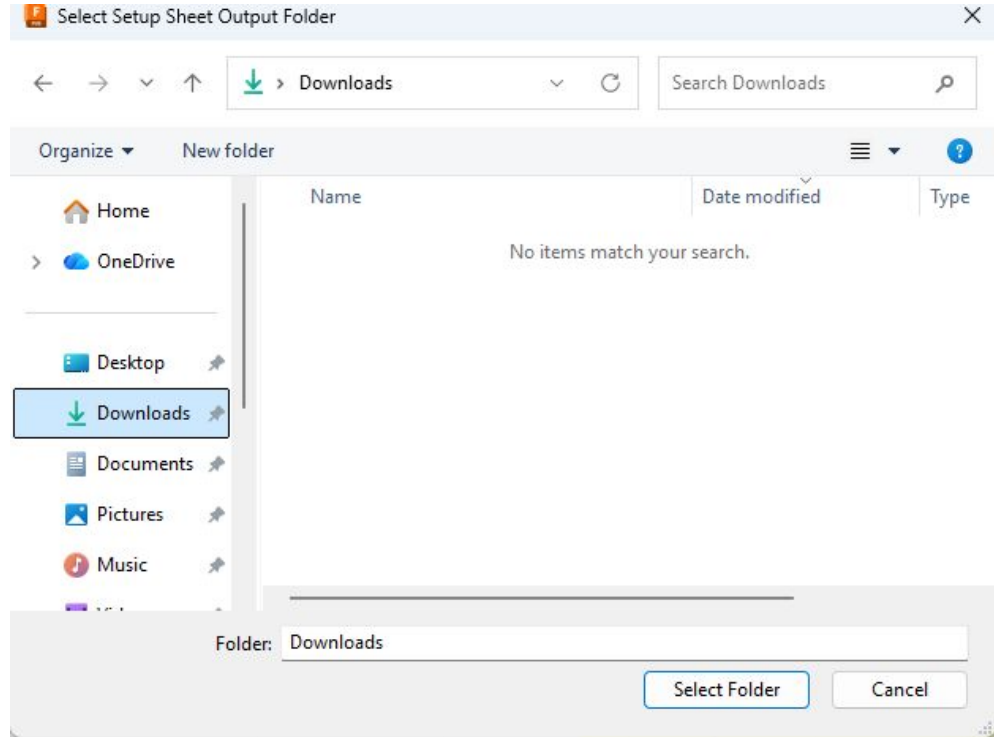
- **Personal Safety:**
 - Required PPE: Safety glasses, hearing protection, close-toed shoes.
 - Prohibited: Loose clothing, untied hair, jewelry, gloves near blade.
- **Pre-Operation Checks:**
 - Ensure your work is not in conflict with work at the adjacent tools
 - Ensure work surface is clear of debris.
 - There must be sufficient material remaining to hold it firmly in place while cutting.
 - Check blade guard is in place and functional.
- **Operation:**
 - Keep hands at least 6 inches from moving parts.
 - Do not leave machine running unattended.
 - Make measurements or adjustments only when machine is stopped.
- **Post-Operation:**
 - Turn off and unplug machine.
 - Clean work area of sawdust and debris.
 - Store tools and materials properly.
 - Ensure blade guard is returned to position.

Fusion 360 - Manufacturing Setup Sheet



Fusion 360 - Manufacturing Setup Sheet

Select your
downloads
folder

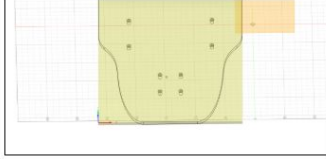


Fusion 360 - Manufacturing Setup Sheet

Rename the
file and save
it to your
Fusion
project

Setup Sheet for Program 1001

JOB DESCRIPTION: Setup2
DOCUMENT PATH: Deck MB v1

Setup	
WCS: #0 Stock: DX: 243.3mm DY: 302mm DZ: 7.35mm Part: DX: 241.3mm DY: 300mm DZ: 8.35mm Stock Lower in WCS #0: X: 0mm Y: 0mm Z: -7.35mm Stock Upper in WCS #0: X: 243.3mm Y: 302mm Z: 0mm	

Total
NUMBER OF OPERATIONS: 2 NUMBER OF TOOLS: 1 TOOL: T1 MAXIMUM Z: 15mm MINIMUM Z: -7.35mm MAXIMUM FEEDRATE: 1000mm/min MAXIMUM SPINDLE SPEED: 15000rpm CUTTING DISTANCE: 1217.78mm RAPID DISTANCE: 727.5mm ESTIMATED CYCLE TIME: 1m:40s

Tools		
T1 D1 L1 Type: Flat end mill DIAMETER: 4mm LENGTH: 10.5mm FLUTES: 3 DESCRIPTION: 04mm Flat Endmill 3F - 1/8" Polycarbonate	MAXIMUM Z: -7.35mm MAXIMUM FEED: 1000mm/min MAXIMUM SPINDLE SPEED: 15000rpm CUTTING DISTANCE: 1217.78mm RAPID DISTANCE: 727.5mm ESTIMATED CYCLE TIME: 1m:34s (86.2%)	HOLDER: Maritool CAT40-ER32-2.35 VENDOR: Maritool PRODUCT: CAT40-ER32-2.35 

Operations		
Operation 1/2 DESCRIPTION: 2D Contour2 STRATEGY: Contour 2D WCS: #0 TOLERANCE: 0.01mm STOCK TO LEAVE: 0mm MAXIMUM STEPDOWN: 3.8mm	MAXIMUM Z: 15mm MINIMUM Z: -7.35mm MAXIMUM SPINDLE SPEED: 15000rpm MAXIMUM FEEDRATE: 1000mm/min CUTTING DISTANCE: 234.31mm RAPID DISTANCE: 995.95mm ESTIMATED CYCLE TIME: 33s (90.3%) COOLANT: Off	T1 D1 L1 Type: Flat end mill DIAMETER: 4mm LENGTH: 10.5mm FLUTES: 3 DESCRIPTION: 04mm Flat Endmill 3F - 1/8" Polycarbonate 
Operation 2/2 DESCRIPTION: 2D Contour3 STRATEGY: Contour 2D WCS: #0 TOLERANCE: 0.01mm STOCK TO LEAVE: 0mm MAXIMUM STEPDOWN: 3.8mm	MAXIMUM Z: 15mm MINIMUM Z: -7.35mm MAXIMUM SPINDLE SPEED: 15000rpm MAXIMUM FEEDRATE: 1000mm/min CUTTING DISTANCE: 993.47mm ESTIMATED CYCLE TIME: 1m:0s (95.7%) COOLANT: Off	T1 D1 L1 Type: Flat end mill DIAMETER: 4mm LENGTH: 10.5mm FLUTES: 3 DESCRIPTION: 04mm Flat Endmill 3F - 1/8" Polycarbonate 